

RESEARCH PROJECT PORTFOLIO

SLR

Optimizing Sign Language Detection with Deep Learning and Preprocessing: A Comparative Study of CNN-Based Models

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 95.6% Accuracy on HGR1

 5 Models • 6 Preprocessing Techniques • 92,000+ Images

01 | PROJECT OVERVIEW

SLR-Bench is a comprehensive benchmarking framework developed to address the persistent challenges in automated Sign Language Recognition (SLR). Real-world sign language images suffer from severe visual degradation: inconsistent lighting, motion blur, low contrast, and cluttered backgrounds. These issues can cause recognition accuracy to drop by up to 40% in suboptimal conditions, preventing reliable deployment of accessibility tools for the deaf and hard-of-hearing community.

This project systematically evaluates five state-of-the-art CNN architectures—**EfficientNet-B0, ResNet-18, VGG19, AlexNet, and a custom CNN**—across six image preprocessing techniques including **CLAHE, gamma correction, contrast stretching, deblurring, and sharpening**. The pipeline was validated on two datasets of increasing scale: a 5,000-image prototype (HGR1) and a full-scale 87,000+ image American Sign Language (ASL) dataset spanning 29 classes. **EfficientNet-B0 with the optimized preprocessing pipeline achieved 95.6% and 94.0% accuracy**, respectively, establishing a reproducible and scalable framework for real-world SLR deployment.

02 | PROJECT DETAILS

Project Type	Research Paper — Comparative Benchmarking Study of CNN Architectures and Preprocessing for Sign Language Recognition
Domain	Computer Vision • Sign Language Recognition • Deep Learning • Accessibility Technology
Publication Status	Presented at ICEEM 2025 — 4th IEEE International Conference on Electronic Engineering, Menoufia University, Egypt (IEEE Xplore)
Core Technologies	Python • PyTorch • EfficientNet-B0 • ResNet-18 • VGG19 • AlexNet • CLAHE • Gamma Correction • Contrast Stretching
Affiliation	MSA University (Egypt) — Faculty of Computer Science

03 | KEY RESULTS & METRICS

UGDA-Net Performance Summary

Model	HGR1 (4 Classes, 5K Images)	ASL (29 Classes, 87K+ Images)
EfficientNet-B0	95.6%	94.0%
ResNet-18	93.05%	91.0%
VGG19	84.7%	84.0%
AlexNet	79.8%	79.0%
Custom CNN	33.5%	30.0%

Impact of Preprocessing Techniques (Average Accuracy Gain)

Technique	Effect
CLAHE (Contrast Limited Adaptive Histogram Equalization)	+7% accuracy gain, best for low-contrast images
Gamma Correction ($\gamma=0.75$)	Significant improvement in underexposed conditions
Contrast Stretching	Enhanced finger contour visibility, especially for shallower models
Gaussian Blur + Unsharp Masking	Moderate but consistent edge definition improvement
Background Removal / Thresholding	Degraded accuracy by 15–20% — excluded from final pipeline

Scalability Insight: The preprocessing pipeline enabled EfficientNet-B0 to maintain 94% accuracy even when scaling from 4 to 29 classes and from 5K to 87K images.

04 | TECHNICAL APPROACH

The Problem

Sign language recognition systems fail in real-world conditions due to severe visual degradation. Inconsistent lighting, motion blur during gesture execution, low contrast between hand and background, and cluttered environments cause standard CNN models to misinterpret hand shapes. Studies report accuracy drops of up to 40% in suboptimal lighting. Additionally, there is no clear consensus on which preprocessing techniques actually improve recognition versus which inadvertently strip away critical spatial features like finger contours and edge boundaries.

The Solution: Optimized Preprocessing + Architecture Benchmarking

This project establishes a reproducible pipeline that systematically pairs six image enhancement techniques with five CNN architectures to identify the optimal combination for scalable, real-world SLR deployment.

Preprocessing Pipeline (6 Techniques)

- **Grayscale Conversion:** Collapses RGB channels into a single intensity map using ITU-R BT.601 luma weighting, reducing dimensionality while preserving structural luminance.
- **Contrast Stretching:** Linear mapping of pixel intensities to the full 0–255 range, improving hand-background separation.
- **Gamma Correction ($\gamma=0.75$):** Non-linear brightening of shadowed regions without overexposing highlights, critical for underexposed images.
- **CLAHE:** Adaptive histogram equalization applied to 8×8 tiles with a clip limit of 2.0, enhancing local contrast while suppressing noise amplification in homogeneous regions.
- **Gaussian Blur + Unsharp Masking:** Noise suppression via Gaussian kernel ($\sigma=1.0$) followed by edge sharpening ($\lambda=1.5$) to accentuate finger boundaries.

Benchmarked Architectures (5 Models)

- **EfficientNet-B0:** Compound scaling across depth, width, and resolution. Achieves highest accuracy with lowest inference latency (~170ms).
- **ResNet-18:** Residual connections mitigate vanishing gradients, enabling deeper feature extraction.
- **VGG19:** Sequential 3×3 convolutions, feature-rich but computationally intensive.
- **AlexNet:** Shallow 5-layer design, CPU-friendly baseline.
- **Custom CNN:** Two convolutional blocks (32, 64 filters) with max-pooling and dense layers.

Training Protocol

- **Input:** 224×224 normalized tensors.
- **Optimizer:** Adam ($\text{lr}=1\text{e-}4$, $\beta_1=0.9$, $\beta_2=0.999$), batch size 32.
- **Epochs:** 5 with early stopping on validation accuracy.
- **Hardware:** CPU-only training — demonstrating feasibility for edge deployment.

Key Exclusion

- **Background Removal & Thresholding** were tested and found to degrade accuracy by 15–20%. The final pipeline intentionally preserves background context, as CNNs learn from lighting gradients and edge shadows as part of the discriminative signal.

05 | KEY CONTRIBUTIONS & IMPACT

Comprehensive Preprocessing Benchmark First systematic evaluation of six preprocessing techniques for SLR, quantifying which methods help (CLAHE, gamma correction) and which hurt (background removal). Provides actionable guidance for future SLR pipeline design.	Optimal Model Identification EfficientNet-B0 emerged as the clear winner, achieving 95.6% on HGR1 and 94.0% on ASL with low inference latency and memory footprint — ideal for edge and mobile accessibility applications.
Scalable Two-Phase Validation Demonstrated pipeline generalizability by scaling from a 4-class prototype (5K images) to full 29-class ASL dataset (87K+ images) with minimal accuracy degradation. Establishes a reproducible framework for real-world deployment.	Social Impact & Accessibility Addresses critical communication barriers for over 70 million deaf and hard-of-hearing individuals globally. The framework provides a foundation for real-time sign language translation tools that can operate on commodity hardware, promoting inclusive education and employment..

06 | ABOUT THE RESEARCHER

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